Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/06/2005
PI Name:	Wang, Ning Ph.D.		
Project Title:	Control of Cell Mechanics and Function by Cell Microenvironment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	nwang@hsph.harvard.edu	**Fax:**	FY 617-432-3468
PI Organization Type:	UNIVERSITY	**Phone:**	617-432-1686
Organization Name:	Harvard Univerity		
PI Address 1:	School of Public Health, Physiology Program		
PI Address 2:	665 Huntington		
PI Web Page:	http://www.hsph.harvard.edu/		
City:	Boston	**State:**	MA
Zip Code:	02115	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	08/01/2001	**End Date:**	10/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Butler, James P. (Harvard University)		
Grant/Contract No.:	NAG 2-1509		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The tasks of this project are: 1) to identify specific transmembrane molecules that transduce mechanical stress into alterations in endothelin-1 gene expression; 2) to determine to what degree endothelin-1 gene expression depends on the stiffness of the extracellular matrix; 3) to determine to what degree cytoskeletal tension and stiffness depends on the stiffness of the extracellular matrix and external mechanical stimulation.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The objective of this project is to elucidate how cell microenvironment controls and regulates cell functions. Elucidating the molecular pathway by which mechanical forces regulate gene expression and determining control of gene expression, cytoskeletal tension, and stiffness by the extracellular matrix mechanics should help us understand the molecular basis of mechanotransduction and mechanoregulation in microgravity as well as in phyiological and pathological conditions.
Task Progress:	Living cells and tissues experience mechanical and chemical stimuli that are critical for their functions. Soluble ligands such as growth factors generally mediate chemical signaling, and immobilized ligands such as extracellular matrix (ECM) proteins mediate both mechanical and chemical signaling across the cell membrane via transmembrane adhesion molecules; it is not, however, clear how a cell responds to mechanical loads applied in different directions via the transmembrane adhesion molecules. One fundamental question is how nuclear structures and functions are influenced by the applied mechanical load at the cell surface. Using the synchronous detection approach, we quantified displacements of nucleolar structures in cultured airway smooth muscle cells in response to a localized physiologic oscillatory load applied via a magnetic bead bound to cell surface integrin receptors. The computed size and shape deformations of nucelolar structures exhibited heterogeneous patterns, with the dominant strain being the two-dimensional bulk strain. The peak displacements of the nucleolus decayed much slower in distance from the load than continuum homogeneous models predict. Increasing the pre-existing tensile stress (prestress) in the cytoskeleton significantly increased the stress propagation efficiency to the nucleolus (defined as nucleolus displacement per surface deformation) whereas decreasing the prestress significantly lowered the stress propagation efficiency to the nucleolus. The latter was supported by the results that decreasing the prestress led to inhibition of long-range stress focusing in the cytoskeleton. Abolishing the stress fibers/actin bundles by plating the cells on poly-L-lysine coated dishes dramatically inhibited stress propagation to the nucleolus. These results demonstrate that the prestress in the cytoskeleton is crucial in mediating stress propagation to the nucleolus, with implications for direct mechanical regulation of nuclear activities and functions (Hu et al., 2005). Our recent experiments have shown that a force of a physiological magnitude, applied via a focal adhesion, can propagate a long distance into the cell. This observation disagrees with existing models that regard the cell as a homogeneous body. To gain insights on the mechanisms of the long-distance propagation of a local force, we developed two models: prestressed string model and stiff bundle model. We show that this “action at a distance” results from the inhomogeneity in the cell: a prestressed and stiff actin bundle guides the propagation of forces over long distances. Our models highlight the enormous ratios of the prestress and the modulus of the actin bundle to the modulus of the cytoskeleton (CSK) network. For a normal cell, the models predict that forces propagate over characteristic lengths comparable to the size of the cell. The characteristic lengths can be altered, however, by treatments of the cell. We provide experimental evidence and discuss biological implications (Wang and Suo, 2005).
Bibliography Type:	Description: (Last Updated: 02/25/2021)
Articles in Peer-reviewed Journals	Hu S, Eberhard L, Chen J, Love JC, Butler JP, Fredberg JJ, Whitesides GM, Wang N. "Mechanical anisotropy of adherent cells probed by a three dimensional magnetic twisting device. " Am. J. Physiol. Cell Physiol. 287: C1184-C1191, 2004. , Nov-2004
Articles in Peer-reviewed Journals	Hu S, Chen J, Wang N. "Cell spreading controls balance of prestress by microtubules and extracellular matrix. " 9: 2177-82, 2004. , Sep-2004
Articles in Peer-reviewed Journals	Wang N, Suo Z. "Long-distance propagation of forces in a cell. " 328: 1133-1138, 2005 , Mar-2005
Articles in Peer-reviewed Journals	Hu S, Chen J, Butler JP, Wang N. "Prestress mediates stress propagation into the nucleus. " Biochem Biophys Res. Commu 329:432-428, 2005. , Apr-2005
Books/Book Chapters	Wang, N. "Control of cell mechanical properties by cytoskeletal prestress" in "Recent Research Developments in Biomechanics" Ed. Pandalai, S.G. Transworld Research Network, 2:1-13 , Jan-2005